

Jatin Nath

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Summary

CS (AI & Data Science) undergraduate at IIIT Kottayam (**CGPA: 7.7**) with a strong foundation in **data structures, algorithms, and object-oriented design**, focused on **backend and systems engineering**. Shipped production-style systems end-to-end: **REST APIs, 3-layer architecture** (Controller/Service/Repository), **task queues** (BullMQ/Redis), **WebSockets**, and **authentication/access-control layers** (JWT, RBAC, audit logging). **Hackathon winner** (550+ teams). Proficient in **Python, C++, TypeScript, Node.js, FastAPI, PostgreSQL, and Docker**.

Experience

Innova Solutions

Software Engineer Intern

On-site

May 2025 – July 2025

- o **GitHub**: <https://github.com/jatinnathh/NeuroFusion> — **Live**: <http://51.21.222.225>
- o Built **RESTful backend** with **RBAC-based access control**, relational data modeling, and **MySQL**-backed storage; resolved **20+ critical bugs** through systematic root-cause analysis; containerized with **Docker** for consistent deployment.
- o Implemented a custom **Stable Diffusion inference pipeline** with asynchronous prompt queuing handling **15+ concurrent GPU requests**; optimized GPU scheduling achieving **40% reduction in idle time** and **40% faster generation** vs. CPU baseline.

Education

Indian Institute of Information Technology Kottayam

B.Tech, Computer Science (AI & Data Science)

2023 – 2027 (Expected)

Projects

DSAgent — Autonomous Backend Platform for Data Pipelines

Feb 2026 – Present

- o **GitHub**: <https://github.com/jatinnathh/DSAgent> — **Live**: <https://dsagentjatin.xyz>
- o Architected a **3-layer backend** (Controller / Service / Repository) with **FastAPI** orchestrating a **6-phase pipeline** (EDA → Cleaning → Visualization → Feature Engineering → Modeling → Evaluation); achieved **12/12 steps, 0 errors**, full pipeline completion in **51s** across **12 orchestrated calls**.
- o Implemented **BullMQ + Redis** task queue for background pipeline execution, email delivery, and audit log flushing — decoupling long-running jobs from the request lifecycle.
- o Designed **complete REST API** with **Clerk + JWT authentication**, input validation, and serialization; implemented **access-control middleware, rate limiting**, and **request timeout** middleware to harden the API surface against abuse.
- o Built **structured logging** (Winston + structlog) with daily rotation, per-request IDs, and a full **audit trail** for traceability; integrated **Elasticsearch** for log indexing with file-based fallback; implemented **circuit breakers** and **retry with exponential backoff** for fault tolerance.
- o Added **graceful shutdown** via FastAPI lifespan hooks to prevent data loss on service restarts/deloys.
- o **Stack**: Node.js, TypeScript, Python, FastAPI, BullMQ, Redis, PostgreSQL, Prisma, Elasticsearch, Winston, structlog.

MediFlow — Hospital Workflow Platform (Winner, TetherX VIT Chennai, 550+ teams)

2026

- o **Live**: <https://turbo-s-final.vercel.app/> — **GitHub**: <https://github.com/jatinnathh/TurboS-Final>
- o Built **Node.js/TypeScript** backend with **complete REST API**; implemented **JWT-based authentication and RBAC authorization** with timestamped audit logs to harden system access control across **5 departments**; data persisted via **PostgreSQL** through Prisma ORM.
- o Engineered a standalone **Socket.IO WebSocket server** for real-time messaging and push notifications; designed a **state-machine workflow engine** (PENDING → IN_PROGRESS → COMPLETED) across **5 workflow types**; integrated Razorpay and third-party APIs.

ModelForge — AutoML Evaluation & Benchmarking Platform

Jul – Aug 2025

- o **Live**: <http://51.21.222.225:8501/> — **GitHub**: <https://github.com/jatinnathh/MLGenie>
- o Deployed on **AWS EC2**; automated benchmarking pipelines reducing manual effort by **70%** across **50+ datasets** and **44+ models** with side-by-side metric tracking dashboards.

Archi — Real-Time Spatial Planning Evaluation via Computer Vision

2026

- o **GitHub**: <https://github.com/jatinnathh/archi.cv> — **Live**: <https://archi-cv.vercel.app>
- o Trained a **custom YOLOv8** model from scratch on a self-collected, manually-annotated dataset of **7 building-type classes** across varied lighting and layout configurations; production threshold tuned at 0.40 confidence, running at **~28 fps**.

Skills

Backend & Systems: Node.js, Express.js, FastAPI, Flask, REST API Design, Controller/Service/Repository Architecture, WebSockets (Socket.IO), Distributed Task Queues

Security-Adjacent: JWT Authentication, RBAC Authorization, Audit Logging, Rate Limiting, Input Validation, Circuit Breakers

Databases: PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM, Relational Schema Design

DevOps & Cloud: Docker, Kubernetes, Jenkins, GitHub Actions, AWS (EC2, S3), Git, Linux, Vercel

Languages: Python, C++, TypeScript, JavaScript, SQL

Achievements & Certifications

2026: Winner (1st)-TetherX Hackathon, VIT Chennai (550+ teams) — Top 10-Hack2Future, IIIT Dharwad (450+ teams)

2024: PG Program in AI & ML — Caltech CTME — IBM Data Science Professional Certificate